

Q1. Transmission Line Analysis [6 Marks]

A transmission line has the following parameters: $Z_0=50 \Omega$, $Z_L=100 \Omega$ (a). Calculate the reflection coefficient (Γ) and the Voltage Standing Wave Ratio (VSWR).

[3 marks]

(b). State whether impedance matching is present or not. Justify your answer. [1 mark]

(c). Calculate the percentage of incident power reflected and transmitted. [2 marks]

(a) Reflection Coefficient (Γ) and VSWR [3 marks]

Reflection Coefficient (Γ):

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \Gamma = \frac{100 - 50}{100 + 50} = \frac{50}{150} = \frac{1}{3} = 0.333$$

Voltage Standing Wave Ratio (VSWR):

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \text{VSWR} = \frac{1 + 0.333}{1 - 0.333} = \frac{1.333}{0.667} \approx 2 \quad \text{Final Answers:}$$

- $\Gamma = 0.333$
- $\text{VSWR} = 2$

(b) Impedance Matching [1 mark]

Impedance matching occurs when:

$$Z_L = Z_0$$

Here:

$$100 \neq 50$$

No impedance matching is present

Justification: Since $Z_L \neq Z_0$, part of the signal is reflected ($\Gamma \neq 0$).

(c) Power Reflected and Transmitted [2 marks]

Reflected Power:

$$\text{Reflected Power} = |\Gamma|^2 = (0.333)^2 = 0.111 \text{ } \rightarrow 11.1\%$$

Transmitted Power:

$$\text{Transmitted Power} = 1 - |\Gamma|^2 = 1 - 0.111 = 0.889 \text{ } \rightarrow 88.9\%$$

Final Answers:

- Reflected Power = **11.1%**
- Transmitted Power = **88.9%**

Final Summary

- $\Gamma = 0.333$
- VSWR = 2
- No impedance matching (since $Z_L \neq Z_0$)
- Reflected power = 11.1%
- Transmitted power = 88.9%

Q2. Electromagnetic Wave Properties [6 Marks]

An electromagnetic wave propagates in free space with an electric field: $E = 8\cos(\omega t - \beta z)$ V/m (a). Write the expression for the average energy density of the EM wave and calculate its value.

[2 marks]

(b). Define the Poynting vector and calculate its average value. [1+2 marks]

(c). Calculate the intensity of the EM wave. [1 mark]

(a) Average Energy Density [2 marks]

Expression:

For EM wave in free space:

$$u_{avg} = \frac{1}{2} \epsilon_0 E_0^2$$

Calculation:

$$u_{avg} = \frac{1}{2} \times 8.854 \times 10^{-12} \times (8)^2 \times \frac{1}{2} \times 8.854 \times 10^{-12} \times 64 \times 32 \times 8.854 \times 10^{-12} u_{avg} = 2.83 \times 10^{-10} \text{ J/m}^3$$

Final Answer:

$$u_{avg} = 2.83 \times 10^{-10} \text{ J/m}^3$$

(b) Poynting Vector [1 + 2 marks]

Definition:

The **Poynting vector** represents power flow per unit area:

$$\vec{S} = \vec{E} \times \vec{H}$$

Average Value:

$$S_{avg} = \frac{E_0^2}{2\eta_0}$$

Calculation:

$$S_{avg} = \frac{8^2}{2 \times 377} = \frac{64}{754} S_{avg} \approx 0.0849 \text{ W/m}^2$$

Final Answer:

- Definition: $\vec{S} = \vec{E} \times \vec{H}$
- Average value:

$$S_{avg} \approx 0.085 \text{ W/m}^2$$

(c) Intensity of EM Wave [1 mark]

Intensity = Average Poynting Vector

$$I = S_{avg}$$

Final Answer:

$$I = 0.085 \text{ W/m}^2$$

Final Summary :

- $u_{avg} = 2.83 \times 10^{-10} \text{ J/m}^3$
- $\vec{S} = \vec{E} \times \vec{H}$
- $S_{avg} = 0.085 \text{ W/m}^2$
- Intensity = 0.085 W/m^2

Q3. Sensor Characterization Parameters [6 Marks]

- (a). Define the quality factor (Q-factor) with its expression. Explain why a high Q-factor is desirable in RF sensors. [2 marks]
- (b). Differentiate between the response time and recovery time of a sensor. [2 marks]
- (c). A sensor has a resonant frequency f_0 . Explain the sensor response when the source frequency is:
- (i) equal to f_0 (ii) slightly higher or lower than f_0
- Comment on the effect on the reflection coefficient S_{11} and the sensor response. [2 marks]

Q3. Sensor Characterization Parameters [6 Marks]

(a) Quality Factor (Q-factor) [2 marks]

Definition:

The **Q-factor** measures the sharpness/selectivity of a sensor's resonance.

Expression:

$$Q = \frac{f_0}{\text{Bandwidth}} = \frac{f_0}{\Delta f}$$

Why is a high Q-factor desirable:

- Produces a **sharp resonance peak**
- Improves **sensitivity** to small changes (e.g., permittivity)
- Enables **better frequency resolution**
- Gives **more accurate detection**

Conclusion: Higher Q → sharper response → better sensing performance

(b) Response Time vs Recovery Time [2 marks]

Parameter	Response Time	Recovery Time
Definition	Time taken to reach steady output after input is applied	Time taken to return to original state after input is removed
When it occurs	During sensing	After sensing
Importance	Determines speed of detection	Determines readiness for next measurement

Key Point:

- Response → **how fast sensor reacts**
- Recovery → **how fast it resets**

(c) Sensor Response vs Frequency [2 marks]

(i) When $f = f_0$ (Resonance):

- Maximum energy transfer occurs
- Sensor shows the **strongest response**
- Reflection coefficient S_{11} is **minimum** (dip)

(ii) When $f \neq f_0$ (slightly higher/lower):

- Sensor moves **away from resonance**
- Response **decreases**
- S_{11} **increases** (more reflection)

Effect Summary:

- At f_0 :
→ Minimum S_{11} , maximum absorption
- Away from f_0 :
→ Higher S_{11} , weaker sensor response

Final Conclusion:

Sensor performance is best at resonance, where reflection is minimum and sensitivity is maximum.

Summary:

- $Q = \frac{f_0}{\Delta f}$, high Q $\rightarrow$ better sensitivity
- Response time: reaction speed; Recovery time: reset speed
- At resonance: $S_{11} \downarrow$, response $\uparrow$
- Away from resonance: $S_{11} \uparrow$, response $\downarrow$

Q4. Transmission Lines in RF Sensors [6 Marks]

- (a). Explain the role of transmission lines in RF sensing systems. Why are simple wires not suitable for RF signal transmission? [2 marks]
- (b). Draw the RLGC equivalent model of a transmission line and write the expression for the characteristic impedance Z_0 . Clearly define all the parameters. Also, state the condition for a lossless transmission line. [2 marks]
- (c). Explain how planar structures such as Interdigital Capacitor (IDC) or Complementary Split Ring Resonator (CSRR) integrated with transmission lines enable RF sensing. [2 marks]

Solution:(a) Role of Transmission Lines in RF Sensing [2 marks]**Role:**

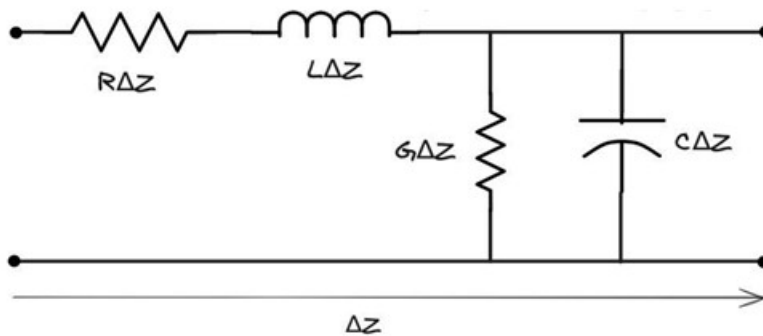
- Transmission lines carry **high-frequency RF signals** from source to sensor and back
- Maintain **signal integrity** (controlled impedance)
- Enable accurate measurement of parameters like S_{11} , **resonance**, **phase**
- Provide **predictable propagation** (voltage/current waves)

Why simple wires are not suitable:

- At RF, wires behave like **distributed elements**, not lumped
- Cause **reflections** due to impedance mismatch
- Lead to **radiation losses and signal distortion**
- No controlled characteristic impedance

Conclusion: Transmission lines are essential for **reliable and accurate RF sensing**

(b)



Primary Constant {

- R = resistance per unit length (Ω/m)
- L = inductance per unit length (H/m)
- G = conductance per unit length (Ω/m or S/m)
- C = capacitance per unit length (F/m)

Characteristic Impedance:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Lossless Condition:

$$R=0, G=0 \Rightarrow Z_0 = \sqrt{\frac{L}{C}}$$

Planar structures like **Interdigital Capacitor (IDC)** and **Complementary Split Ring Resonator (CSRR)** are integrated with transmission lines to create **resonant RF sensors**.

- These structures introduce **localized electric fields** and **resonance** in the transmission line.
- When a material under test (MUT) is placed near them, it alters the **effective permittivity**.

- This causes a **shift in resonant frequency** and a change in the reflection coefficient S_{11} .
- **IDC** mainly provides strong **capacitive coupling (fringing fields)** → sensitive to surface materials.
- **CSRR** produces a **sharp resonance (high Q)** → highly sensitive detection.

Q5. S-Parameters and RF Measurement [6 Marks]

- (a). Why S-parameters are preferred at high frequencies? [2 marks]
 (b). Explain the physical meaning of S_{11} and S_{21} . [2 marks]
 (c). A device has $S_{11} = -20$ dB. Interpret this value in terms of the percentage of power reflected and impedance matching quality. [2 marks]

(a) Why S-parameters are preferred at high frequencies? [2 marks]

- At high frequencies, **voltage and current are difficult to measure directly**
- RF circuits exhibit **wave behavior (reflections, standing waves)**
- S-parameters use **incident and reflected power waves**, which are easier to measure using instruments like VNA
- They naturally account for **impedance mismatch and reflections**

Conclusion: S-parameters are preferred because they are **practical, accurate, and suited for high-frequency RF analysis**

(b) Physical meaning of S_{11} and S_{21} [2 marks]

- S_{11} (**Input Reflection Coefficient**):
 - Ratio of reflected to incident signal at input
 - Indicates **impedance matching**
 - Lower S_{11} → better matching
-
- S_{21} (**Forward Transmission Coefficient**):
 - Ratio of output to input signal

- Indicates **how much signal is transmitted through the device**
- Higher $S_{21} \rightarrow$ better transmission

Summary:

- $S_{11} \rightarrow$ reflection / matching
- $S_{21} \rightarrow$ transmission / gain or loss

(c) Given $S_{11} = -20$ dB [2 marks]

Step 1: Convert dB to magnitude

$$|S_{11}| = 10^{\frac{-20}{20}} = 10^{-1} = 0.1$$

Step 2: Power reflected

$$\text{Reflected Power} = |S_{11}|^2 = (0.1)^2 = 0.01 = 1\%$$

Step 3: Interpretation

- Only **1% power is reflected**
- **99% power is transmitted**

Impedance Matching Quality:

- Very **good impedance matching**
- Reflection is very low
- Suitable for efficient RF systems

Final Answer:

- Reflected power = **1%**
- Matching = **excellent (well matched system)**

Section B

Solution Q1

(a) Quality Factor (Q) [2 marks]

Formula:

$$Q = \frac{f_r}{\text{Bandwidth}}$$

Sensor A:

$$Q_A = \frac{2.4 \text{ GHz}}{120 \text{ MHz}} = \frac{2400}{120} = 20$$

Sensor B:

$$Q_B = \frac{2.4 \text{ GHz}}{60 \text{ MHz}} = \frac{2400}{60} = 40$$

Comment:

- **Sensor B has higher Q (40)** → sharper resonance
- Hence, **Sensor B has better selectivity and sensitivity**

(b) Sensitivity [2 marks]

Formula:

$$\text{Sensitivity} = \frac{\Delta f}{\Delta \varepsilon_r}$$

Sensor A:

$$S_A = \frac{15}{0.5} = 30 \text{ MHz/unit}$$

Sensor B:

$$S_B = \frac{30}{0.5} = 60 \text{ MHz/unit}$$

Comparison:

- Sensor B has **higher sensitivity**
- Detects changes more effectively

(c) Impedance Matching using S_{11} [2 marks]

- Sensor A: $S_{11} = -8 \text{ dB}$
- Sensor B: $S_{11} = -14 \text{ dB}$

Interpretation:

- More negative $S_{11} \rightarrow$ **less reflection** $\rightarrow$ **better matching**

Conclusion:

- **Sensor B has better impedance matching**
- Because it reflects **less power** compared to Sensor A

(d) Definitions [2 marks]

Return Loss (RL):

$$RL = -20 \log_{10} |S_{11}|$$

- Measures **power reflected back**
- Higher RL $\rightarrow$ **better matching**

Insertion Loss (IL):

$$IL = -20 \log_{10} |S_{21}|$$

- Measures **power loss through the device**
- Lower IL → **better transmission**

(e) Identify Sensor Structures [2 marks]

- **Sensor A:**
Interdigital Capacitor (IDC)
- **Sensor B:**
Complementary Split Ring Resonator (CSRR)

Final Summary

- $Q_A = 20$, $Q_B = 40 \Rightarrow B$ better
- Sensitivity: $A = 30$, $B = 60 \Rightarrow B$ better
- S_{11} : B (-14 dB) → better matching
- RL → reflection, IL → transmission loss
- A = IDC, B = CSRR

Q2 PLz see ppt for derivation

PPT 9

Q3 (a) plz see ppt 10

(b) Given:

$$Z_{\text{source}} = 50 \, \Omega, Z_L = 100 \, \Omega$$

Formula:

For a quarter-wave transformer:

$$Z_t = \sqrt{Z_{\text{source}} \cdot Z_L}$$

Calculation:

$$Z_t = \sqrt{50 \times 100} = \sqrt{5000} Z_t \approx 70.7 \, \Omega$$

Final Answer:

$$Z_t \approx 70.7 \, \Omega$$

Section C

Section C Answers (10 × 1 = 10 Marks)

Q1:

Insertion Loss (dB) = -10 log₁₀(P_{in}/P_{out})

$$\text{Insertion Loss} = 10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right) = 10 \log_{10} \left(\frac{10}{5} \right) = 10 \log_{10} (2) \approx 3 \text{ dB}$$

Answer: (B) 3 dB

Q2:

Low conductivity → **less attenuation** → **deeper penetration**

Answer: (B) Increases

Q3:

For $\lambda/2$ line:

$$Z_{in} = Z_L$$

Answer: (C) Equal to load impedance

Q4:

Quarter-wave transformer:

- High $\leftrightarrow$ Low transformation
Answer: (C)

Q5:

$S_{21}=0 \rightarrow$ no signal passes

Answer: (B) No transmission

Q6:

Ideal VSWR:

1:1

Answer: (B)

Q7:

Intrinsic impedance:

$$Z = \frac{E}{H}$$

Answer: (B)

Q8:

Skin depth definition:

$$\frac{1}{e}$$

Answer: (C)

Q9:

In poor conductor:

- Skin depth **not dependent on frequency for poor conductors**
Answer: (C)

Q10:

SRR works on:

- **LC resonance**
Answer: (C)